

Aminci Jonathan Adebajo Gana

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🔗 portfolio-website-zeta-eight-47.vercel.app/ 🌐 github.com/HilbertSpaceExplorer

Summary

Mechanical Engineering BSc. student at Leibniz Universität Hannover with hands-on experience in robotics, computer vision, sensor integration, and ML-based sensor data analysis. Seeking working student roles in machine learning, robotics, computer vision, and AI-based industrial systems.

Availability: 15–20 h/week during semester; full-time during semester breaks

Technical Skills

Programming

Python, MATLAB/Simulink, Arduino C/C++, Java basics, JavaScript/TypeScript basics

Machine Learning

scikit-learn, PyTorch basics, TensorFlow basics, classification, regression, model evaluation, data preprocessing

Computer Vision

OpenCV, camera calibration, stereo vision, ArUco/AprilTag detection, pose estimation, image processing

Tools

Git/GitHub, VS Code, Linux/Ubuntu, CAD, Microsoft Office

Web Development

HTML, CSS, React/Next.js basics, Vercel deployment

Languages

English

Native

German

C1

Mandarin

HSK2/A2

Spanish

A1/A2

Yoruba

B2 comprehension

Experience

Student Assistant, Institute of Mechatronic Systems

04/2024 – Present

- Supported research and student projects in semi-automated driving, computer vision, and mechatronic systems.
- Developed and tested Python/OpenCV-based perception workflows using ArUco/AprilTag markers, stereo vision, and Intel RealSense cameras.
- Assisted with robotics lab experiments, including sensor integration, data acquisition, calibration, debugging, and evaluation.

Internship, Körber Feinmechanik GmbH

02/2023 – 04/2023

Gained practical experience in precision mechanics, manufacturing processes, and technical workshop workflows.

Digital Marketing, Cerdana

05/2021 – 03/2022

Managed online marketing activities and customer communication for a small business.

Internship, Autocleanfix Repair Services Ltd.

12/2020 – 03/2022

Supported vehicle repair and maintenance tasks, developing practical mechanical troubleshooting skills.

Education

Mechanical Engineering B.Sc., Leibniz Universität Hannover

04/2023 – Present | Hannover, Germany

Expected graduation: 2026

Feststellungsprüfung, Niedersächsisches Studienkolleg

04/2022 – 02/2023 | Hannover, Germany

Final grade: 1.5

High School Diploma (WASSCE), Model Secondary School Maitama

11/2020 | Abuja, Nigeria

Final grade: High School Diploma (WASSCE) - 3 Distinctions & 6 Credits

German equivalent grade: 2.1

Projects

Continuum Robot Modeling Under External Forces , Bachelor Thesis

Bachelor thesis project focused on modeling continuum robot shape under external contact forces using tendon load cells, base force sensing, and stereo vision.

Tools: MATLAB, Simulink, Python, ROS, Intel RealSense, OpenCV

ML-Based State Classification and Flow Estimation in a Compressed-Air System

Applied classification and regression methods, including scikit-learn-based models, to compressed-air sensor data for operating-state detection and airflow estimation.

Tools: Python, scikit-learn, pandas, NumPy, classification, regression, sensor data analysis

Vision-Based Robustness Evaluation of PID Distance Control ,

Python, OpenCV, NumPy, PID control, data analysis

Developed a computer vision-based workflow to evaluate PID distance-control robustness using distance error, overshoot, oscillation, and settling behavior.

Tools: Python, OpenCV, PID Control, Data Analysis

Computer Vision Laptime Measurement for Line-Following Robots

Built a video-based stopwatch system to automatically detect start/finish-line crossings and measure robot lap times, reducing manual timing errors during line-following robot tests.

Tools: Python, OpenCV, Video Processing

Volunteering & Leadership

Teaching & Organization of Online Mathematics Class 2020

Awareness Programmes 2019

Server of Breakfast with Our Neighbours 2019

Volunteer at Orphanage 2017